

Customer Retail SQL Analysis

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1 CREATE DATABASE customer_retail_db;
2
3 USE customer_retail_db;
4
5 # 'Customer Retail' is a cleaned & transformed 'Customer shopping pattern' dataset,
  after performing EDA using Python.
6 # Now will generate important business insights & KPIs using SQL queries:
7
8 # Performing total 19 'Ad Hoc' Business Queries:
9
10 # Q1.Sales Performance: What is the 'Total revenue' and the total number of
   transactions recorded in the dataset?
11 # Ans: Revenue generated is 2,33,200 USD with a record of 3902 Total Orders
12
13 SELECT
14     SUM(amt_usd) AS Total_Revenue,
15     COUNT(cust_id) AS Total_Orders
16 FROM customer_dbs;
17
18 # Q2.Revenue breakdown: Which product category is driving most revenue, and how many
   items were sold in each?
19 # Ans: Category 'Clothing' is driving most revenue with 1,04,383 USD
20
21 SELECT category,
22     SUM(amt_usd) AS category_rev,
23     COUNT(*) AS items_sold
24 FROM customer_dbs
25 GROUP BY category
26 ORDER BY category_rev DESC;
27
28 # Q3.What are the top 3 most purchased products within each category? - using CTE
   Functions
29
30 WITH item_counts AS (
31     SELECT category, item_purchased,
32         COUNT(cust_id) AS total_orders,
33     ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(cust_id) DESC) AS
   item_rank
34 FROM customer_dbs
35 GROUP BY category, item_purchased
36 )
37 SELECT item_rank,category, item_purchased, total_orders
38 FROM item_counts
39 WHERE item_rank <=3;
40
41 # Q4.Revenue by Gender & Avg Transactions (ATV): Revenue generated by male vs. female
   customers?
42
43 SELECT gender,
44     SUM(amt_usd) AS total_spend,
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45         round(avg(amt_usd),2) AS avg_transaction_value
46 FROM customer_dbs
47 GROUP BY gender;
48
49 # Q5.High-Spending Discount Users: Identify customers who used discounts but still
    spent above the avg purchase amount?
50 -- using subquery
51
52 SELECT cust_id, gender, amt_usd
53 from customer_dbs
54 where discount_applied = 'Yes'
55 AND amt_usd >=
56 (select avg(amt_usd) from customer_dbs);
57
58 # Q6.Group Analysis: Purchases made by each age-group, total_rev & which group has the
    highest average rating?
59 # Ans: 'Young Adult' has the max purchases & highest avg_rating of 3.8
60
61 SELECT age_group,
62         COUNT(*) AS total_purchase,
63         ROUND(AVG(rating), 2) AS avg_rating,
64         sum(amt_usd) as total_rev
65 FROM customer_dbs
66 GROUP BY age_group
67 ORDER BY total_purchase DESC;
68
69 # Q7.Top 5 Products by Rating: Find products with the highest average review ratings?
70
71 select item_purchased, round(avg(rating),2) as avg_rating
72 from customer_dbs
73 group by item_purchased
74 order by avg_rating desc
75 limit 5;
76
77 # Q8.Shipping Comparison: What is the avg purchase amounts between Standard and
    Express shipping?
78
79 select shipping_type,
80         round(avg(amt_usd),2) as avg_amt
81 from customer_dbs
82 where shipping_type IN ('Express', 'Standard')
83 group by shipping_type;
84
85 # Q9.Do subscribers spend more: Compare avg spend & total revenue across subscription
    & Non subcr status?
86 # Ans: NO, 73% revenue comes from Non-subscribers
87
88 SELECT subscription,
89         COUNT(cust_id) AS total_customers,
90         ROUND(AVG(amt_usd),2) AS avg_spend,
91         ROUND(SUM(amt_usd),2) AS total_rev
92 FROM customer_dbs
93 GROUP BY subscription
94 ORDER BY total_rev, avg_spend DESC;

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95
96 # Q10.Does having a subscription impact the customers average rating (or) their total
97 # number of previous purchases?
98 # Ans: NO, it doesn't effects
99
100 SELECT subscription,
101        COUNT(*) AS cust_count,
102        round(AVG(rating),2) AS Avg_Rating,
103        AVG(prev_purchase) AS Avg_Prev_Purchases
104 FROM customer_dbs
105 GROUP BY subscription;
106
107 # Q11.Are customers who are repeat buyers (more than 5 previous purchases) also likely
108 # to subscribe?
109 # Ans: NO
110
111 SELECT subscription, COUNT(cust_id) AS repeat_buyers
112 FROM customer_dbs
113 WHERE prev_purchase > 5
114 GROUP BY subscription;
115
116 # Q12.Which are the top 5 locations (states/regions) in terms of total 'Sales'
117 # revenue?
118 # Ans: Top 5 locations are - Montana, California, Illinois, Idaho, Nevada
119
120 SELECT location, SUM(amt_usd) AS Total_Sales
121 FROM customer_dbs
122 GROUP BY location
123 ORDER BY Total_Sales DESC
124 LIMIT 5;
125
126 # Q13.Discount-Dependent Products: Identify 5 products with the highest percentage of
127 # discounted purchases?
128
129 select item_purchased,
130        round(100 * sum(CASE WHEN discount_applied = 'Yes' THEN 1 else 0 END)/count(*),2) as
131        discount_rate
132 from customer_dbs
133 group by item_purchased
134 order by discount_rate desc
135 LIMIT 5;
136
137 # Q14.Seasonal Trends: Which season experiences the highest average purchase amount?
138 # Ans: Fall season
139
140 SELECT season, round(AVG(amt_usd),2) AS Avg_Spend
141 FROM customer_dbs
142 GROUP BY season
143 ORDER BY Avg_Spend DESC;
144
145 # Q15.What is the most frequently used payment method among customers? - PayPal
146
147 SELECT payment_method, COUNT(*) AS Usage_Count
148 FROM customer_dbs

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144 GROUP BY payment_method
145 ORDER BY Usage_Count DESC;
146
147 # Q16.Customer Segmentation: Classify customers into New, Returning, and Loyal
    segments based on purchase history?
148 -- Using CTE Functions
149
150 with customer_type as (
151 SELECT cust_id, prev_purchase,
152 CASE
153     WHEN prev_purchase = 1 THEN 'New'
154     WHEN prev_purchase BETWEEN 2 AND 10 THEN 'Returning'
155     ELSE 'Loyal'
156     END AS customer_segment FROM customer_dbs)
157
158 select customer_segment, count(*) AS 'No of Customers'
159 from customer_type
160 group by customer_segment;
161
162 # Q17.Loyalty & Frequency: What is the avg "previous purchases" count for customers
    grouped by their purchase frequency?
163 -- (e.g., Weekly, Monthly)?
164
165 SELECT frequency_of_purchases,
166     round(AVG(prev_purchase),2) AS Avg_Loyalty_Score
167 FROM customer_dbs
168 GROUP BY frequency_of_purchases
169 ORDER BY Avg_Loyalty_Score DESC;
170
171 # Q18. How does applying a discount affect the average amount spent per transaction?
172 # Ans: Discount applied seems to have 14% low transactions compared to 'No Discount'
    category
173
174 SELECT discount_applied, AVG(amt_usd) AS Avg_Spend,
175     COUNT(*) AS Total_Transactions
176 FROM customer_dbs
177 GROUP BY discount_applied;
178
179 # Q19.Preferred Shipping by Demographic: Which shipping type is most popular among
    "Young Adult" vs "Senior" customers?
180
181 SELECT age_group,
182     shipping_type,
183     COUNT(*) AS Count
184 FROM customer_dbs
185 WHERE age_group IN ('Young Adult', 'Senior')
186 GROUP BY age_group, shipping_type
187 ORDER BY age_group, Count DESC;
188
189
190
191

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